

Motor Imagery and Direct Brain–Computer Communication

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Invited Paper

Motor imagery can modify the neuronal activity in the primary sensorimotor areas in a very similar way as observable with a real executed movement. One part of EEG-based brain–computer interfaces (BCI) is based on the recording and classification of circumscribed and transient EEG changes during different types of motor imagery such as, e.g., imagination of left-hand, right-hand, or foot movement. Features such as, e.g., band power or adaptive autoregressive parameters are either extracted in bipolar EEG recordings overlaying sensorimotor areas or from an array of electrodes located over central and neighboring areas. For the classification of the features, linear discrimination analysis and neural networks is used. Characteristic for the Graz BCI is that a classifier is set up in a learning session and updated after one or more sessions with online feedback using the procedure of “rapid prototyping.” As a result, a discrimination of two brain states (e.g., left- versus right-hand movement imagination) can be reached within only a few days of training. At this time, a tetraplegic patient is able to operate an EEG-based control of a hand orthosis with nearly 100% classification accuracy by mental imagination of specific motor commands.

Keywords—Brain–computer interface (BCI), EEG classification, event-related desynchronization (ERD), event-related synchronization (ERS), motor imagery.

I. A SHORT OVERVIEW OF EEG-BASED BCI SYSTEMS

EEG-based brain–computer interfaces (BCI) can be realized in an externally (stimulus)-paced mode (synchronous BCI) or in an internally paced mode (asynchronous BCI). In the former case, specific mental states have to be generated in response to an external event that means changes of

brain activity occur in a predefined time window. In the latter case, the subject is free to intend a mental state or a specific thought. BCI systems operating in an externally paced mode can use different types of brain signals as input. On the one side, oscillatory EEG activity [1], [2] and slow cortical potential shifts [3] are analyzed and classified. On the other side, various types of event-related potentials are used as, e.g., VEP [4], P300 [5], or steady-state VEPs [6].

Self-regulated slow cortical potential (SCP) shifts in the EEG are investigated by Birbaumer’s group in Tübingen to operate a thought translation device (TTD). The SCPs are measured in a 2-s window and referred to a 2-s baseline and used to move a ball-like light into a target. Patients already using this system are able to write text after many training sessions [3].

An interesting approach to the communication system introduced by Farwell and Donchin [7] is based on the presentation of a 6×6 letter matrix, whereby in short intervals one of the rows or one of the columns of the matrix is flashed. Focussation of the user to one letter/item elicits an EP component called P300. With this system, a communication rate of about seven items/min can be obtained [5]. Another approach to a BCI is to evaluate the amplitude of steady-state VEPs. When, for example, two items on the monitor are flickering with slightly different frequencies in the alpha/beta range, the steady-state VEP amplitude is enhanced, when the participants shift their gaze to the corresponding light-source [6]. The implantation of a special electrode into the outer layer of the neocortex allows to record multiunit activity [8]. This activity in form of spikes is transformed into a pulse train and used, e.g., for cursor control and selection of letters/items on a monitor. Spelling rates of about three letters/min are reported with such patients.

The BCIs of Wolpaw’s group in Albany and the Graz group are both based on motor imagery and classification of sensorimotor EEG rhythms and are therefore discussed in more detail. In the 1980s, Wolpaw *et al.* [2] started on EEG-based cursor control in normal adults using band

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power centered at 9 Hz. The EEG was recorded bipolarly from the left sensorimotor area and the subject was instructed to move a cursor up or down into a target, whereby upward movement required large and downward movement low mu rhythm amplitude. The amplitude of sensorimotor EEG rhythm was translated directly into cursor movement. At this time, the BCI-system in Albany [9] uses an autoregressive (AR) model applied to short overlapped EEG segments. From the AR spectrum, band power is calculated in 4–5-Hz-wide frequency bands and the power sum is used as independent variable in a linear equation that defines the cursor movement. This means the cursor moves (upward or downward) as a linear function of the EEG amplitude in a specific frequency band. Cursor control can be improved when two EEG channels are used, one over the sensorimotor area of each hemisphere referenced to a common average reference, and band power calculated in the most reactive frequency bands [10].

As early as 1996, the Graz group [11] reported on an EEG-based BCI able to discriminate between three brain states, where the subject was instructed to execute a specific movement (right hand, left hand, right foot) in the first three sessions and to imagine the same movement in the next 1–2 sessions. Band power (5–35 Hz) was estimated in intervals of 250 ms from three bipolar EEG channels. Four power estimates of each of the three channels were concatenated to form 12-dimensional vectors that were presented to a neuronal network-based classifier. The nonlinear classifier was trained with examples from the first sessions without feedback. The online results from the motor imagery sessions with feedback varied around 50% (worst case of classification accuracy is 33.3%).

The next important step in the Graz BCI was to select the most reactive frequency bands for each subject by using the distinction sensitive learning vector quantization (DSLQV) [12]. DSLQV is an improved learning vector quantizer (LVQ, [13]) and a valuable tool for automatic feature selection that can be applied to assess, e.g., the most relevant frequency components and/or the most relevant electrode positions. For the selected bands (normally two bands per channel are found), the band power is calculated and classified by a nonlinear classifier (LVQ). Subject-specific frequency band selection raised the online classification accuracy by almost 10% [14]. Details of the Graz BCI system based on motor imagery are discussed in the next sections.

II. NEUROPHYSIOLOGICAL CONSIDERATIONS

In this section, a short comment on the dynamics of brain oscillations is made and results on motor imagery are briefly reviewed.

A. Dynamics of Brain Oscillations

Populations of neurons can form complex networks whereby feedback loops are responsible for the generation of oscillatory activity. In general, the frequency of such oscillations becomes slower with increasing number of synchronized neuronal assemblies [15]. Two types of os-

cillations have importance for the BCI: the Rolandic mu rhythm in the range 7–13 Hz and the central beta rhythm above 13 Hz, both originating in the sensorimotor cortex [16], [17]. Sensory stimulation, motor behavior, and mental imagery can change the functional connectivity within the cortex and results in an amplitude suppression [event-related desynchronization (ERD)] or in an amplitude enhancement [event-related synchronization (ERS)] of mu and central beta rhythms [18], [19]. These changes can be, e.g., due to modulating influences of neurochemical brain systems, changes in the strength of synaptic interactions, or changes of intrinsic membrane properties of the local neurons. It is expected that in the control of the cortical dynamics of brain oscillations, the interplay between thalamic relay cells and the thalamic reticular nucleus is of importance [20].

The dynamics of brain oscillations associated with sensory and cognitive processing and motor behavior can form complex spatiotemporal patterns. So, e.g., a synchronization of higher frequency components embedded in a desynchronization of lower frequency components can be found on a specific electrode location at the same moment of time (see Fig. 1, upper part). Simultaneous desynchronization and synchronization of 10-Hz components are possible on different scalp locations (see Fig. 1, lower part).

Preparation and planning of self-paced hand movement results in a short-lasting desynchronization (ERD) of Rolandic mu and central beta rhythms. First reports on this phenomenon from Jasper and Penfield [16] and Gastaut [17] were confirmed by Pfurtscheller and Aranibar [18], who proposed a quantification of the ERD. More recently, Toro *et al.* [21] and Crone *et al.* [22] reported about electrocorticographic (ECoG) recordings and ERD in the alpha band associated with hand and foot movement. The general finding is that similar to the mu rhythm (around 10 Hz), beta oscillations desynchronized during the preparation and execution of a motor act. However, alpha and beta frequency components differ with respect to temporal behavior. Fig. 1 (upper panel) presents data from a voluntary movement experiment [23] demonstrating the existence of at least three different types of oscillations at the same electrode location over the sensorimotor hand area during brisk finger lifting. The alpha band (mu) ERD (10–12 Hz) starts 2.5 s before movement-onset, reaches maximal desynchronization shortly after movement-onset, and recovers to baseline level within a few seconds. The central beta activity (14–18 Hz), in contrast, displays a short-lasting ERD during initiation of movement followed by a synchronization (ERS) with a maximum after movement execution. It is of interest that the beta ERS occurs while the mu rhythm is still attenuated. One important feature of these beta oscillations is their strict somatotopical organization in MEG [24] and EEG [25]. The 40-Hz oscillations (36–40 Hz) reveal a sharp power increase shortly before movement-onset. Such movement-related gamma oscillations, however, are rarely found in the human EEG.

B. Motor Imagery

Motor imagery may be seen as mental rehearsal of a motor act without any overt motor output. It is broadly accepted

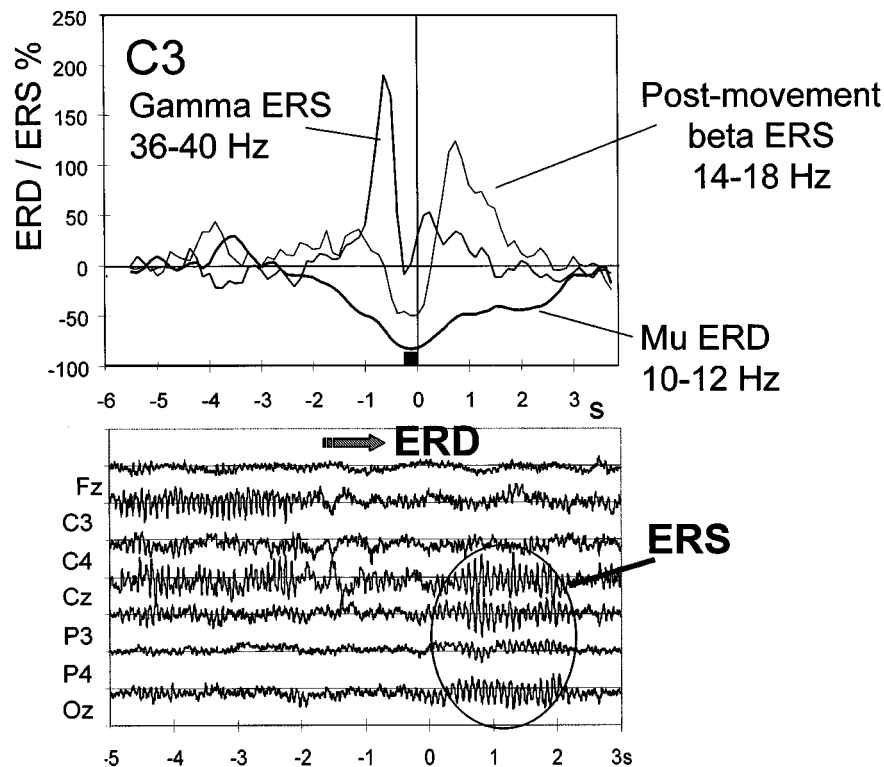


Fig. 1. Upper panel: Superimposed band power time courses computed for three different frequency bands (10–12 Hz, 14–18 Hz, and 36–40 Hz) from EEG trials recorded from electrode position C3 during right index finger lifting. EEG data are triggered with respect to movement-offset (vertical line at $t = 0$ s). Note the beginning of mu ERD around 2.5 s prior to movement-offset, the maximum of gamma ERS immediately prior to movement-onset and the maximum of the beta ERS within the first second after movement-offset. Lower panel: Examples of ongoing EEG recorded during right finger movement. Movement-onset at $t = 0$ s. Note the EEG desynchronization (ERD) at central electrode locations prior to movement-onset and the enhanced alpha band activity over the posterior region (ERS) during movement (modified from [19]).

that mental imagination of movements involves similar brain regions/functions which are involved in programming and preparing such movements [26]. According to this view, the main difference between performance and imagery is that in the latter case execution would be blocked at some cortico-spinal level [27].

Functional brain imaging studies monitoring changes in regional cerebral blood flow (rCBF) revealed indeed similar patterns of activity during motor imagery and actual movement performance. An increase of the rCBF has mainly been located in the supplementary motor area during imagination of sequential finger movements [28]. Moreover, recent positron emission tomography (PET) [27] and functional magnetic resonance imaging (fMRI) studies [29] revealed activation of a number of cortical and subcortical areas, including, e.g., the premotor cortex, the anterior cingulate gyrus, superior and inferior parietal areas, and the cerebellum. Thus, activation has been observed in various structures involved in the early stage of motor control (i.e., motor programming), but not in the primary sensorimotor cortex. More recent fMRI studies, in contrast, detected some activation in the primary motor cortex during motor imagery, though to a lesser extent than during actual motor performance [30]. Increased motor cortex activation during motor imagery has been supported by studies using

transcranial magnetic stimulation in showing an increase of motor responses during mental imagination of movements [31].

Several EEG studies further confirm the notion that motor imagery can activate primary sensorimotor areas [32]–[34]. A blocking of the central mu rhythm with motor imagery was reported in early clinical EEG observations [17]. Similar cortical activity over the contralateral hand area during execution and imagination of hand movement has further been found with dc potential measurements [32] and based on dipole source analysis of electric and magnetic fields [33]. Furthermore, recent high-resolution EEG experiments of our group [34], [35] showed that independent of the required motor task, imagination versus overt execution of a given movement, the most prominent EEG changes were localized over the corresponding primary sensorimotor cortex. During the imagination of a right-hand or left-hand movement, for example, we found a similar ERD over the contralateral hand area as is usually found during planning or preparation of a real movement. This imagination-related ERD shows different time courses in the alpha and beta bands (Fig. 2, upper panel). Similar to the self-paced movement task, the ERD in the beta band shows a fast recovery and is followed by a short-lasting synchronization (beta ERS). Examples for the localization of beta ERS mapped

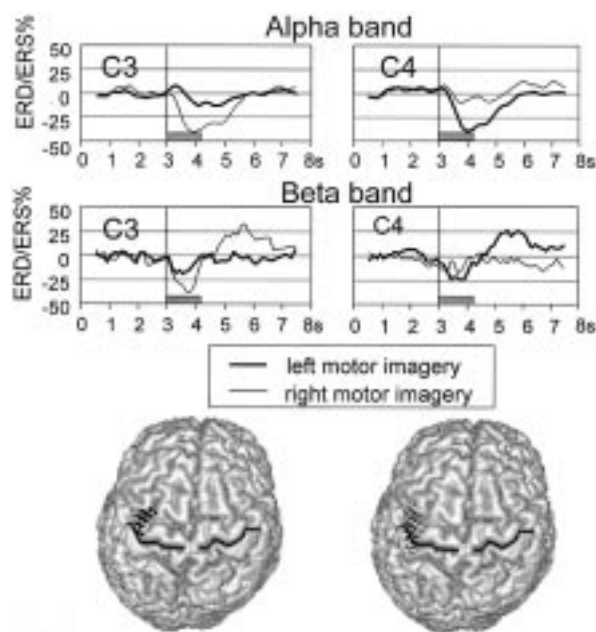


Fig. 2. Upper part: Grand average ERD/ERS curves recorded over left and right sensorimotor cortex during motor imagery. The ERD/ERS time courses were calculated for the selected bands in the alpha (upper panels, $n = 16$) and beta range (lower panel, $n = 8$). Positive and negative deflections, with respect to the baseline, represent a band power increase (ERS) and decrease (ERD), respectively. A gray bar indicates the time period of cue presentation. Lower part: Spatial distribution of the beta ERS calculated for the cortical surface of one subject after execution of a real right hand movement (left) and imagination of the same movement (right). This model was constructed from 200 T1-weighted transversal MR images acquired with a 1.5-tesla Philips Magnetron system. The three-dimensional electrode positions, measured with a magnetic digitizer (Polhemus), were transformed into the MRI coordinate system. The central sulcus is marked by a thin black line. Relative strengths of dipolar moments computed using a realistically shaped three-shell boundary element head model, not the linear estimation method, are indicated by black disks (modified from [34]).

on the reconstructed cortical surface of one representative subject (Fig. 2, lower panel) illustrate a focus close to the primary hand area after both a real executed and an imagined right-hand movement. This observation is in line with recent neuromagnetic studies suggesting that the central 20-Hz activity mainly originates in the primary sensorimotor cortex [24] and, moreover, documents the supposed similarity of neural circuitry involved in mental representation and movement execution.

It is of interest, that during overt execution of the movement, the initially contralateral ERD develops a bilateral distribution [19], [21], whereas during mental simulation this ERD remains mostly limited to the contralateral hemisphere. This means that the suppression of mu and central beta rhythms is more pronounced at the contralateral hemisphere when subjects imagine one-sided hand movements [35] than when they actually perform such movements. This observation led us to utilize motor imagery as control strategy to achieve asymmetrical electrocortical responses and to use, e.g., left-right differences in the sensorimotor EEG to provide a control option in one dimension [35], [36].

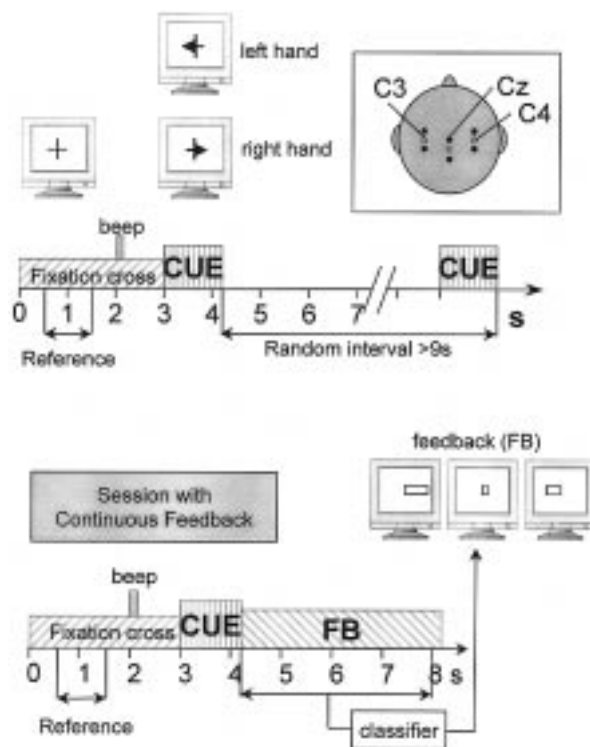


Fig. 3. Upper panel: Imagination paradigm without feedback. The cue stimulus in form of an arrow indicates the type of imagination. The reference period used for calculation of ERD curves is indicated. Insert: electrode positions for three bipolar EEG-channels placed over hand and foot areas. Lower panel: Imagination paradigm with continuous feedback. The subject's task is to extend the feedback bar in the indicated direction. EEG is classified online for a period of 4 s.

III. COMPONENTS OF GRAZ BCI

A. Experimental Paradigm

In the standard paradigm for the discrimination of two mental states, the experimental task is to imagine either right-hand or left-hand movement depending on a visually presented cue stimulus. The subject fixates on a computer monitor 150 cm in front of her/him. Each trial is 8 s long and starts with the presentation of a fixation cross at the center of the monitor, followed by a short warning tone (beep) at 2000 ms. At 3000 ms, the fixation cross is overlaid with an arrow at the center of the monitor for 1250 ms, pointing either to the left or to the right (Fig. 3, upper part). Depending on the direction of the arrow, the subject is instructed to imagine, e.g., a movement of the left or the right hand.

Two different types of feedback are used: 1) discrete delayed feedback and 2) continuous feedback. Discrete feedback consists of a symbol presented in the center of the monitor at 6000 ms: the type of symbol (large or small “+” or “-” or “0”) depends on how well a subject-specific classifier can distinguish the two EEG patterns related to left versus right motor imagery in a fixed time window from 3250 to 4250 ms. In the case of continuous feedback, the imagination task is controlled by means of a horizontal feedback bar. Depending on the cue stimulus, the subject's task is to extend a bar toward the right or left boundary of the monitor

by mental imagination of moving the right or left hand, respectively. The bar appears at time 4250 ms and is presented over a 4-s period (Fig. 3, lower part). The length of the bar directly corresponds to the linear distance function obtained by online analysis of the EEG signals [36].

Normally, each session consists of four experimental runs of 40 trials (20 “left” and 20 “right” trials) and lasts for about 1 h. The sequence of “left” and “right” trials, as well as the duration of the breaks between consecutive trials (ranging between 500 and 2500 ms), is randomized throughout each experimental run.

B. Parameter Estimation and Classification

1) *Band Power Estimates and Nonlinear Classification*: The EEG is composed of different types of oscillatory activities whereby oscillations in the alpha and beta band are particularly important to discriminate between different brain states during visual and motor imagery. One possibility to select parameters from the ongoing EEG is first to calculate short-term power spectra in intervals of, e.g., 250 ms and thereafter estimate the band power [9]. Another is to first bandpass-filter the EEG and then to estimate the band power in intervals of, e.g., 250 ms thereafter [14], [37]. Both methods assume stationarity of the EEG over short time intervals. The selection of most reactive frequency can be done by the DSLVQ algorithm [12]. Online classification of two motor imagery tasks with band power and a neuronal network (LVQ) revealed an average accuracy of 78% in four subjects [37].

2) *Adaptive Autoregressive Model and Linear Discrimination*: A classical approach for estimating time-varying AR parameter is the segmentation based approach [38]. In this case, the data is divided into short segments and the AR parameters are estimated from each segment. The result is a time course of the AR parameters that describes the time-varying characteristics of the process. The segment length determines the accuracy of the estimated parameters and defines the resolution in time. The shorter the segment length, the higher is the time resolution but this has the disadvantage of an increasing error of the AR estimates.

Alternatively, the adaptive autoregressive (AAR) algorithms can perform calculation concurrent to the data acquisition, where no buffering is required and the model has the following form:

$$y_k = a_{1,k}y_k + \dots + a_{p,k}y_{k-p} + \varepsilon_k$$

where y_k is EEG time series, $a_{1,k} \dots a_{p,k}$ are time-variant autoregressive parameters, p is the model order and ε_k is a noise process.

For AAR estimation, the least mean squares (LMS) approach [39] and the recursive least squares (RLS) approach [40] are possible besides other methods. The calculation requires low computational effort and no buffer memory. For this reason, the algorithms are well suited for online analysis.

The AAR parameters $\hat{\mathbf{a}}_t^{(i)}$ are estimated for every sample time point for the EEG channels from, e.g., electrode positions $C3$ and $C4$. This results in a feature vector $\mathbf{d}_t^{(i)}$ with

a dimension of $2 \cdot p$ for every trial i and for every sampling point t

$$\mathbf{d}_t^{(i)} = \begin{bmatrix} \hat{\mathbf{a}}_t^{(i)}, & C3 \\ \hat{\mathbf{a}}_t^{(i)}, & C4 \end{bmatrix}.$$

With a linear discriminant analysis (LDA) between the two classes L and R (e.g., left or right motor imagery), a weight vector $\mathbf{w}_t$ and offset w_0 are found with which the distance D is computed

$$D_t^{(i)} = \mathbf{w}_t^T \cdot \mathbf{d}_t^{(i)} - w_0.$$

D_t is a one-dimensional time-varying function that can be calculated for every single trial. $D_t^{(i)}$ is called the time-varying signed distance function (TSD). A distance $D_t^{(i)} > 0$ and $D_t^{(i)} < 0$ means that trial i is classified as a trial belonging to class 1 or 2, respectively. This procedure is repeated for every classification time point t , and a time course of the error rate ERR_t is obtained [40].

The AAR method in combination with LDA was applied successfully in the work on healthy subjects to discriminate different motor imagery patterns [37], in a patient to control a hand orthosis by motor imagery [41] and in a field study on 324 subjects (details on this field study are discussed later).

3) *Common Spatial Patterns and Linear Discrimination*: Frequently, the ERD/ERS focus during motor imagery is not located beneath electrode positions $C3$ and $C4$. One possibility to improve classification accuracy in these circumstances is to use the method of common spatial patterns (CSP), which constructs spatial filters that are optimal for discrimination of two populations of EEG. The method is based on the simultaneous diagonalization of two matrices [42]. The method was recently applied to the classification of movement related trials and motor imagery EEG [43], [44].

For the analysis, the EEG is bandpass filtered in the relevant frequency band and each trial is represented as $N \times T$ matrix E , where N is the number of channels (i.e., recording electrodes) and T is the number of samples. The normalized spatial covariance can be calculated as

$$C = \frac{EE^T}{\text{trace}(EE^T)}.$$

After averaging the spatial covariance of the two populations (i.e., left and right motor imagery), the composite spatial covariance is given as $C_c = \bar{C}_l + \bar{C}_r$. Then, C_c can be factored as $C_c = U_c \lambda_c U_c^T$, where U_c are the eigenvectors of C_c and λ_c are the corresponding eigenvalues. The variances in the space spanned by U_c can be equalized by the whitening transformation

$$P = \sqrt{\lambda_c^{-1}} U_c^T.$$

It can be shown that if $\bar{C}_l(\bar{C}_r)$ are transformed as $S_l = P\bar{C}_lP^T$ ($S_r = P\bar{C}_rP^T$), then S_l and S_r share common eigenvectors, i.e., if $S_l = B\lambda_l B^T$, then $S_r = B\lambda_r B^T$ and $\lambda_l + \lambda_r = I$. The projection of whitened EEG onto the first

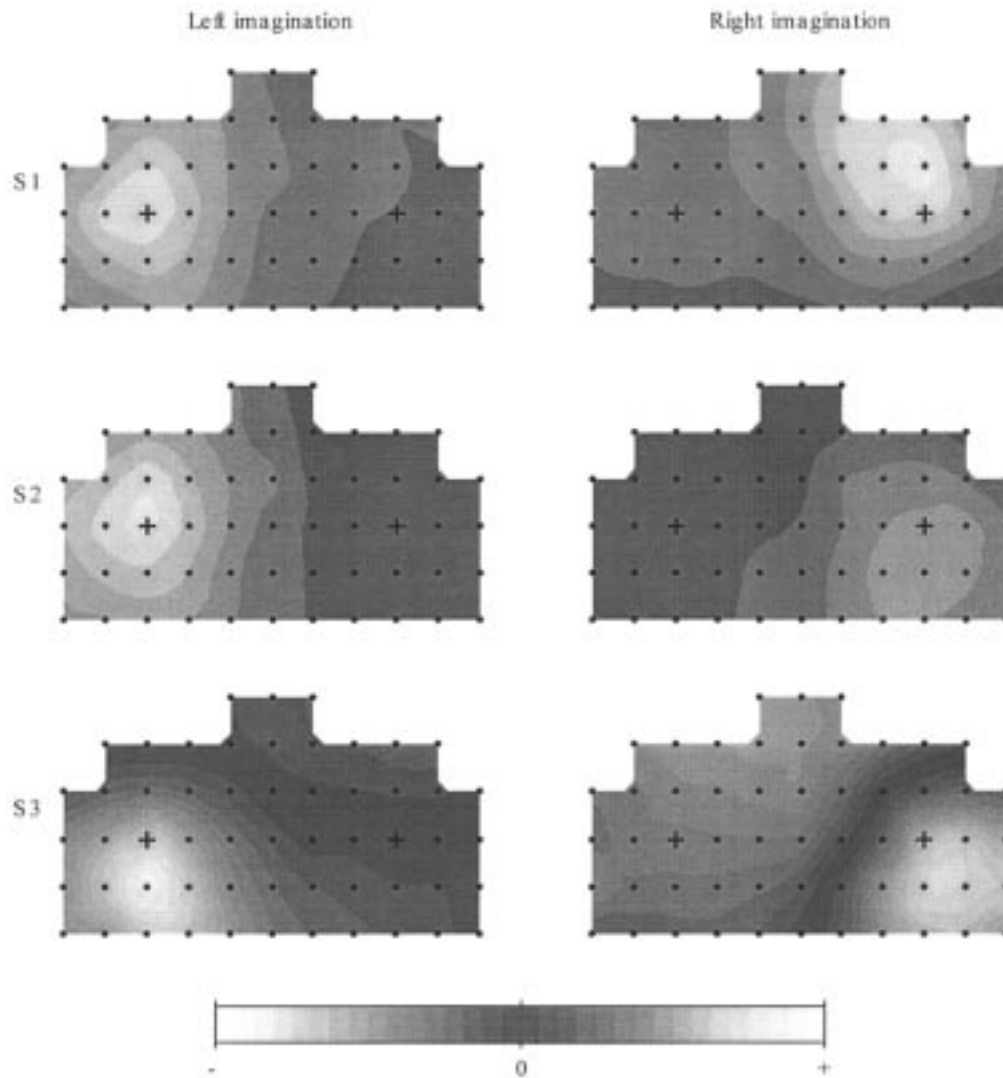


Fig. 4. Most relevant common spatial patterns of three subjects. The left (right) column shows the pattern most suited for detection of left (right) hand motor imagery where light colors represent relevant regions. Electrode positions are marked with a dot except for electrodes *C3* and *C4*, which are marked with a “+.” The contour plots are obtained with cubic interpolation between the CSP values calculated for each electrode (modified from [44]).

and last eigenvectors in B gives feature vectors which are optimal for discriminating the two populations in a least squares sense.

With the projection matrix $W = (B^T P)^T$ the mapping of a trial E is given as $Z = WE$. The columns of W^{-1} are the common spatial patterns and can be seen as time-invariant EEG source distribution vectors. The features for classification are obtained by mapping the EEG and retaining only a small number of signals most suitable for discrimination. Due to the construction of Z , these are the m first and last rows of Z . The variances of these m signals are log-transformed to normalize the distributions and linear discriminant analysis is sufficient for discrimination of the two populations.

Fig. 4 displays examples of the most relevant common spatial patterns for three subjects [44]. Left motor imagery causes relatively increased EEG over the left hemisphere because on the contralateral hemisphere event-related desyn-

chronization of EEG takes place. This behavior is reflected in large coefficients for electrodes on the left hemisphere. The most important patterns show their strongest modulation at electrodes above the motor cortex. However, for none of the subjects was the focus exactly centered at electrode positions *C3* and *C4*, which are used for EEG classification in most BCI experiments. The CSP method can be effectively used in real-time on standard PC software. The online classification accuracy for three healthy subjects in a two-imagery task experiment ranged from 87%–98% [45].

4) *Hidden Markov Model:* The two classification methods based on autoregressive parameters or common spatial patterns classify the ongoing EEG in short windows where stationarity is assumed. Dynamic EEG changes such as, for example, patterns of desynchronization cannot be modeled, and are therefore not considered for classification. To overcome the problem of nonstationarity, the hidden Markov models (HMMs) were introduced for modeling the

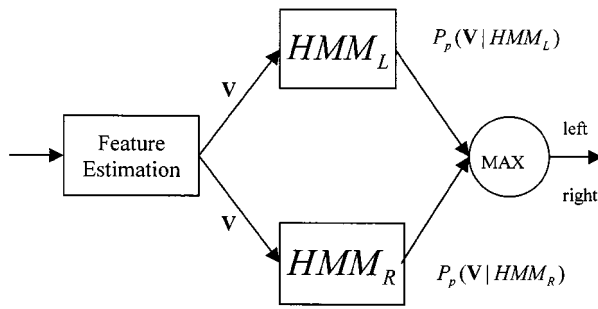


Fig. 5. BCI system comprising the feature estimation and two HMM's used for classification.

dynamic EEG changes. An HMM describes a first-order time domain process which is defined as a process where the conditional probability of the current event merely depends on the most recent event. The HMM itself could be seen as a finite automate containing discrete states, emitting a feature vector at every time point. This feature vector is modeled using Gaussian mixtures. The transition probabilities from one state to the other state are described using a transition matrix [46].

The BCI-system comprises a feature estimation and an HMM for every type of motor imagery (see Fig. 5). Classification is based on the selection of maximum best path probability [46] for the feature sequence $\mathbf{V}$ given the HMMs (HMM_L for the left and HMM_R for the right motor imagery, respectively). During the feedback session, the length of the feedback bar is calculated proportional to the difference of $P_p(\mathbf{V}|HMM_L)$ and $P_p(\mathbf{V}|HMM_R)$. The HMM trained with Hjorth parameters estimated in two bipolar derivates was used for online classification of EEG patterns related to left- versus right-hand imagery. The online classification rate occurring in four healthy subjects varied between 75% and 95% [47].

C. Hardware–Software Requirements and Rapid Prototyping

The recent Graz BCI consists of an IBM-compatible Pentium II PC operating at 233 MHz and an RTI800a data acquisition board (Analog Devices, Norwood, MA), with 32 single ended analog input channels, eight digital inputs, and eight digital outputs. The system has three ISA slots; a maximum of 96 analog channels can be used. The analog to digital converter (ADC) has a resolution of 12 bits. For further hardware details, see [48].

The PC, with Windows 95 installed, is equipped with a real-time Kernel expansion. Simulink is used for the calculation of different parameters, which describe the current state of the EEG in real-time, while Matlab handles the data acquisition, timing, and presentation of the experimental paradigm.

A protocol for “rapid prototyping” was introduced consisting of seven steps [48]:

- Step 1) **Selection of parameter estimation and classification algorithm**
- Step 2) **Implementation of the parameter estimation and classification algorithms**

Algorithms can be programmed in Matlab or a block diagram can be developed using Simulink (an example is given in Fig. 6). Simulink’s graphical user interface enables the user to build block diagrams using drag-and-drop techniques and provides the ability to write special blocks for online analysis, called S-functions (system-functions).

Step 3) **Offline simulations and tests of the Simulink block diagrams**

Step 4) **Connection of the Simulink model to the real world**

Real-time programs communicate with external input–output (I/O) devices via a device driver that contains the necessary code to interface Simulink to the RTI800a DAQ board.

Step 5) **Real-time code generation**

Once the desired results are achieved with Simulink offline tests (Step 3), the real-time C code is directly generated, compiled, linked, and downloaded to a real-time kernel with the real-time workshop (RTW).

Step 6) **Communication with the real-time program**

Simulink in the external mode can be used as a graphical front end to the corresponding model. When the model is downloaded to the kernel, it can be started from Simulink and runs in real-time under Windows. An interprocess communication channel connects the real-time process to the Simulink block diagram. With the Matlab Application Program Interface (Matlab API) and the Simulink External Interface, it is possible to interact with the real-time program without stopping the execution.

Step 7) **Real-time tests**

It might be necessary to go back to Step 1 to adjust the algorithm.

IV. MAN–MACHINE LEARNING DILEMMA (MMLD)

Two types of experimental sessions are used in the Graz BCI: 1) training sessions where data are collected to set up a subject-specific classifier (Fig. 7, upper panel) and where no feedback is provided and 2) test sessions where the classifier is used to classify the subject’s EEG online while the imagination is requested. In this session, feedback is given to the subject (Fig. 7, lower panel).

It should be expected that in BCI experiments, where feedback is given to the subject, the classification accuracy improves with increasing number of sessions. This has, however, not always happened, as shown in a BCI experiment and discrimination between left/right hand movement imagination, where no significant increase above 80% was found with increasing number of feedback sessions [14]. This may be explained by the man–machine learning dilemma (MMLD). MMLD implies that two systems (man and machine) are strongly interdependent but have to be adapted independently. The starting point of this adaptation

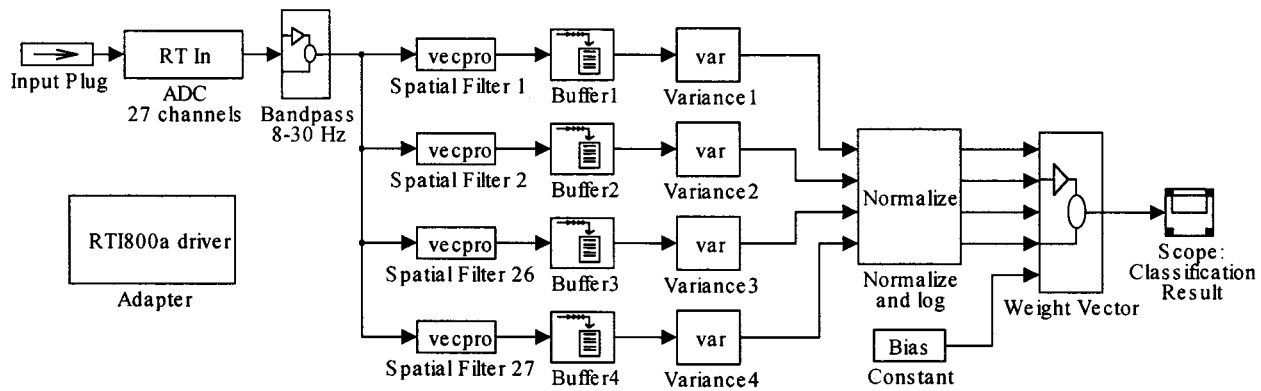


Fig. 6. Simulink model for the real-time analysis of the EEG. A device driver for the RTI800a (DAQ board of Analog Devices) makes the connection to the real world. In this case, the input block represents analog input channels 1 to 28 (EEG#1 to EEG#27, Trigger). Channels 1 to 27 are bandpass filtered between 8 and 30 Hz. The output signal is then passed to the two most (Spatial Filter 1 and Spatial Filter 27) and two second most (Spatial Filter 2 and Spatial Filter 26) discriminating common spatial filters. After temporal and spatial filtering, the variances of the resulting four time series were calculated for a one second window, normalized and also log-transformed. The resulting features were classified with the weight vector (WV). This result was used to control the feedback bar on the monitor (modified from [45]).

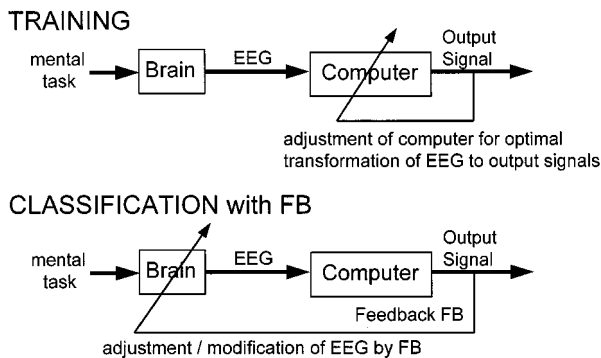


Fig. 7. Schematic display of the two adaptive systems involved in brain computer interactions. Upper panel: The computer learns to recognize different brain states. Lower panel: Feedback to the subject modifies brain activity.

is the training of a “machine” to recognize certain EEG patterns of a subject. During this phase, no feedback is given. As soon as feedback is provided, each feedback results in an adaptation of man to machine: man tries to repeat success and avoid failure. Wrong feedback can elicit frustration, a response likely to be associated with a widespread EEG desynchronization, whereas a correct feedback might lead to a reinforcement of the specific EEG patterns. In both cases, the feedback could introduce noise and deteriorate the classification performance. Besides the direct effect of feedback, the visually presented feedback stimulus itself (bar, cursor, letter, etc.) also can modify the EEG and also introduce some noise to the system. Changing distributions of EEG patterns require adaptation of pattern recognition methods, i.e., adaptation of machine to man. This, again, influences the system’s performance responses and, via feedback, influences the behavior of man and results in further variations of the EEG patterns.

The experiments give evidence that adaptation of the machine (computer) to the man (brain) is necessary. One solution is to adapt the computer after one or multiple sessions.

For this purpose, the above-mentioned protocol for rapid prototyping was developed. This means that not only various types of parameter estimation methods and classification algorithms can be implemented very fast [48], but also a classifier can be updated within minutes after each session with or without feedback. Such a procedure of “rapid prototyping” used for the Graz BCI with 27 electrodes placed over sensorimotor areas and right/left motor imagery is illustrated in Fig. 8.

For the parameter estimation, the method of common spatial patterns (CSP) and for discrimination, the linear discriminant with weight vectors (WV) was used. Besides the CSP, the WV also were updated in the course of six sessions. This procedure was tested in three subjects [45] and resulted in a classification accuracy of about 86%, 94%, and 98% within only three days (see Fig. 9).

V. CAN A VISUAL TARGET STIMULUS MODIFY THE EEG IN SENSORIMOTOR AREAS?

From experiments with EEG-based cursor movement, we have evidence that already 500 ms after the visual target appearance a target-specific response is found in recordings from electrode positions overlaying the hand area [4]. This early response raises the question whether the visual cue used in the motor imagery paradigm activates sensorimotor areas. From animal experiments it is known that in an instructed delay period with a visual cue, the motor cortex neurons begin to code the direction of the upcoming movement as soon as 100 ms after visual cue onset [49]. Further, it is demonstrated that visual input associated with recognition of objects results in an increased excitability of motor cortex neurons [50].

We have used the method of common spatial patterns (CSP) [43], [44] to study the time course of classification accuracy in a movement imagination experiment with a visual cue stimulus [51]. The classification procedure was

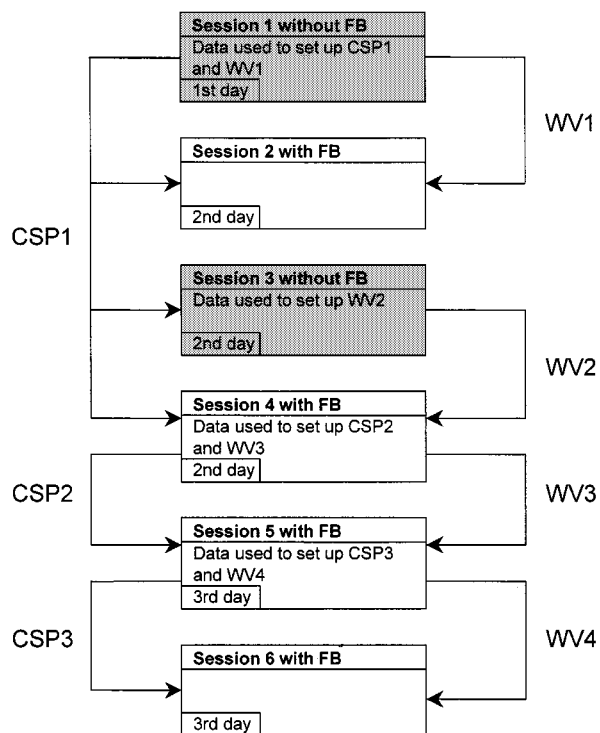


Fig. 8. Flowchart of six BCI sessions with and without (gray boxes) feedback (FB) for one subject. Altogether, three common spatial filters (CSP) and four weight vectors (WV) were set up. The sessions were performed within three days (modified from [45]).

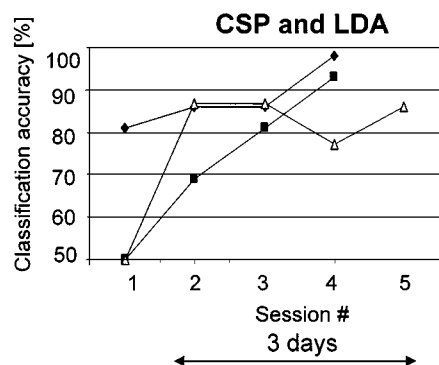


Fig. 9. Improvement of the online classification accuracy over three days in three subjects with common spatial patterns (CSP) and linear discriminant analysis (LDA).

performed in single trials for time segments of 250-ms length, overlapping by 125 ms, starting 3 s prior to cue stimulus onset and lasting 8 s. A correct discrimination between EEG trials associated with left/right hand movement imagination started 250–500 ms after cue-onset in the four investigated subjects.

An example of a spatially filtered time series from one subject is displayed in Fig. 10. It can be seen that: 1) the response is found as early as 250 ms after cue onset and 2) the responses are different depending on the visual cue (arrows pointing to the left and right, respectively). The fast increase of classification accuracy can be interpreted as a result of the visually cue triggered process in the primary sensorimotor cortex, which might be sustained due to the imagination process itself. Thus, two different processes are hypothe-

sized, one probably unconscious but demonstrable in central EEG and the other conscious due to imagination of the movement. It is important to note that only observations of a specific visual cue stimulus might selectively activate the right or left primary sensorimotor cortex some 100 ms after stimulus onset. This is also supported by experiments with subjects responding to laterally presented letters with right and left index finger movement, where the visual cue induced an early increase of negativity contralateral to the relevant stimulus with a maximum of 200–300 ms after cue-onset [52].

This finding, that two different visual targets can result in two distinguishable spatiotemporal EEG patterns as early as 250 ms after stimulus-onset, is of special importance in BCI research and opens new ways to install a fast communication channel between brain and computer.

VI. HOW MANY PEOPLE ARE ABLE TO OPERATE AN EEG-BASED BCI?

A group of 324 visitors of an exposition (Steiermärkische Landesausstellung 2000) participated voluntarily in a BCI field study. The participants were aged between eight and 90 years and spent around 30 min for the BCI investigations including electrode montage. The experimental procedure was divided into two sessions. The first session consisted of 40 trials without feedback. Thereafter, a subject specific linear classifier was set up to give feedback to the subject. During the second session, the subject had the task of controlling a horizontal bar on the computer screen (again, 40 trials). About half of the subjects were instructed to imagine either left or right hand movement after a visual cue; the other half was instructed to imagine either hand or foot movement. Two bipolar derivations were mounted over the right- and left-hand areas and over the hand and foot representation areas, respectively. For the parameter estimation, the AAR algorithm or the band power method was used. In order to test the separability of the EEG patterns in the first session by a linear discriminant, a 10×10 -fold cross validation procedure was performed.

It is of interest that in about 12% of the subjects, the two brain states related to motor imagery were distinguishable with an accuracy of better than 80% after only about 10 min of training. Further, 78% of the subjects were classified with accuracy between 60%–80% and only in 10% no discrimination between brain states was achieved. The subjects of the group with 60%–80% classification accuracy can be seen as good candidates for whom a further improvement is expected in additional training sessions.

VII. CONTROL OF ORTHOSIS IN A TETRAPLEGIC PATIENT

Here, we report data from a tetraplegic patient (T.S., 22 years old, male) of level C4/C5. His high-level spinal cord injury has resulted in complete paralysis of lower and upper limbs, except for residual muscle activity of the left biceps. The goal of the pilot project was to restore the hand grasp function of his left hand with an electronic hand orthosis. The control of the orthosis should be performed with the Graz BCI system. The project included the development of

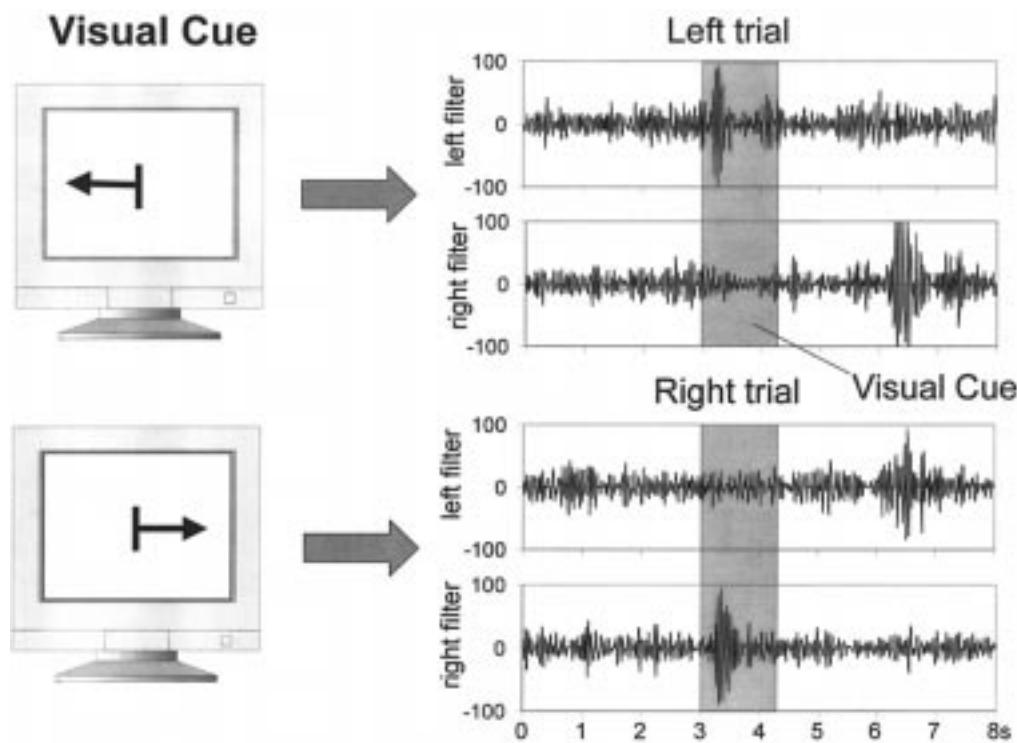


Fig. 10. Examples of spatially filtered time series for one subject from filters calculated for the time window 3.5–4.5 s. From top to bottom: time series from left (upper panels) and right (lower panels) imagination obtained with “left filter” and “right filter.” Y-scale in arbitrary units, X-scale time in seconds. Visual cue presentation is marked by “gray” (modified from [51]).

an electronic hand orthosis supporting the grasp function of the left hand and EEG feedback training to gain self-control of electric brain oscillations.

The patient has been trained for four months to produce cue-stimulus-dependent changes in his Rolandic mu and beta rhythms, using the motor imagery task described above. According to the visual cue-stimulus on the computer screen, the patient should imagine one of two different types of upper or lower limb movements (e.g., left versus right hand, right hand versus both feet) during the following 4-s period. A representative training day consisted of 2–4 sessions, each of which comprised 160 trials and lasted about 60 min.

During the feedback period, the subject was required to concentrate on motor imagery to specifically modify the Rolandic oscillations. The EEG was recorded from two bipolar channels over the corresponding representation area (according to the instructions left hand, right hand, foot) close to the electrode positions $C4$, $C3$, and Cz , respectively. The EEG signals were filtered between 0.5 and 30 Hz and sampled at 128 Hz. The data were analyzed by an AAR model with order 6 and classified by LDA.

Results of AAR and LDA over 62 sessions with and without feedback are shown in Fig. 11. Feedback was either given with a horizontal bar or with the hand orthosis. During sessions 1–28, the subject imagined left- versus right-hand movement and achieved a classification accuracy of about 65%. In the following sessions (29–53), various mental strategies were tested (involving, e.g., left- and right-sided hand versus foot movement) and the classification improved slightly to an average of about 75%. With the beginning

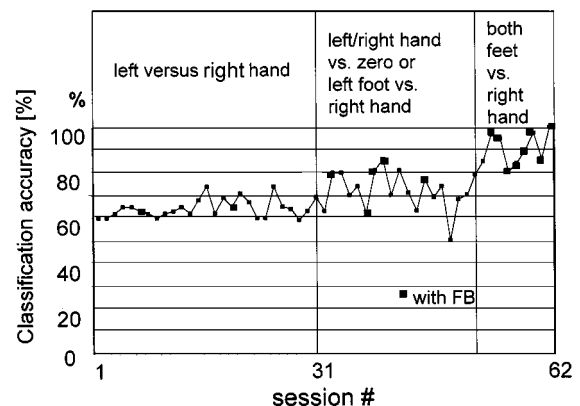


Fig. 11. Improvement of classification accuracy over a time span of four months. (□) indicate sessions with feedback. T. S. started with a classification accuracy of about 65% and ended after four months with an accuracy of nearly 100% (modified from [41]).

of session 54, the strategy was to imagine a movement of both feet versus right hand and the classification accuracy improved sharply to about 95%. In the last session, even 100% were reached, meaning the two EEG-patterns related to different types of motor imagery were online correctly classified in all 160 trials [41].

The most interesting point is the increase of the accuracy after changing the strategies of motor imagery. We found that the classification of imagination of both feet movement versus right-hand movement was superior to all other control modalities. Inspection of the EEG data revealed that the subject had learned after 53 sessions to induce a 16-Hz rhythm

by feet imagery localized close to the midcentral foot representation area, which can easily be detected and, hence, used for the control of the hand orthosis.

VIII. CONCLUSION

At this time, nearly all BCI systems operate in a synchronous mode and are therefore externally paced. They use as input signal either oscillatory EEG components [1], [2], [9], [11], slow cortical potential shifts [3], evoked potential components [5], [7], or steady-state EPs [6]. Asynchronous BCI systems need no type of external timing and analyze the EEG signal continuously. First results on movement-related potential detection are reported by Levine [53] and Birch and Mason [54]. Whether such asynchronous BCI systems are also able to detect mere mental events without any motor response has to be proved.

The information transfer rate defined in bits/trial or bits/selection depends strongly on the classification accuracy. So, e.g., in a four-class BCI (four different brain states), 24 bits/min (12 trials/min) are theoretically possible. In the case of a classification accuracy of 80%, the rate drops to about 12 bits/min [55]. This example shows that a high classification accuracy is critical for good performance. An increase of accuracy is possible by an increase of the signal-to-noise ratio (SNR) whereby the signal can consist of brain oscillations in specific frequency bands, slow cortical potential shifts, or different types of evoked potentials. Different types of temporal or spatial filters can reduce the noise and thereby enhance the SNR. Another way for maximizing the SNR is the use of implanted electrodes [8], [56].

A good classifier is also of crucial importance for satisfactory classification results. Besides linear equation [2], [9], linear discrimination analyses [37], [45], and neural networks [11], [12] support vector machinery (SVM) have gained increased attention. SVMs are less susceptible to overfiltering problems than classical feedforward networks, and can therefore be used with relatively smaller sizes of training samples [57].

Shortening of trial length in synchronous BCI lower than 5 s is possible when only faster oscillations, as, e.g., beta rhythms are used for communication purposes. The alpha (μ) rhythm needs some seconds to recover from the desynchronized state, whereas the beta rhythm recovers much faster within 1 s [19], [23]. In the case of measuring SCPs' shifts, intervals of 2 s and baseline intervals of 2 s are already very close to the limit and only minor shortening will be possible.

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